

DEPARTMENT OF ENGINEERING CYBERNETICS

TTK4265 - OPTICAL REMOTE SENSING

Pushbroom Spectrometer Corrections

Assignment Description

Length max 10 pages
Deadline: 4 October 2026 13:30 AM

1 Introduction and Background

The use of HyperSpectral Imagers (HSIs) is increasing in many fields. Optical remote sensing is one of these fields. Before hyperspectral instruments, custom designed multispectral instruments were used that record a specific set of bands tuned to a specific application. Hyperspectral instruments on the other hand are general use, but as a downside, generate much more data than multispectral instruments. Before a given HSI can be used in a mission or field test, laboratory calibration and characterization of the instrument is needed to ensure proper data quality, and to identify any potential issues in the final assembled configuration.

A thorough calibration campaign can be both time consuming and expensive to perform, but in this lab exercise, you will perform a smaller calibration campaign focused on measuring a set of key performance indicators for HSIs, spectral, radiometric, and dark current characterizations.

You will be working with a miniaturized HSI build at NTNU from mass produced (commercial-off-the-shelf (COTS)) optical components and a 3D printed frame Sigernes et al. 2018. The assignment is largely based on the work found in Henriksen, Prentice et al. 2022; Henriksen, Sigernes et al. 2022. The laboratory facilities, equipment and time are limited. To prevent introducing delays in the assignment, spend some time prior to your lab hours to prepare properly.

1.1 Instrument

An exploded view of the HSI can be seen in Fig. 1. The original designed aimed for a wavelength range of 400nm to 700nm, but the full sensor reaches a larger spectral range. The full range is not usable due to transmission and absorptions properties of the lenses blocking light below 400 nm, low Quantum Efficiency (QE) of the CMOS sensor outside the 400 nm to 700 nm range, and second order diffraction effects in the grating that appear above 800 nm. Thus This assignment focuses on the calibration and characterization in the spectral range between 400 and 700 nm.

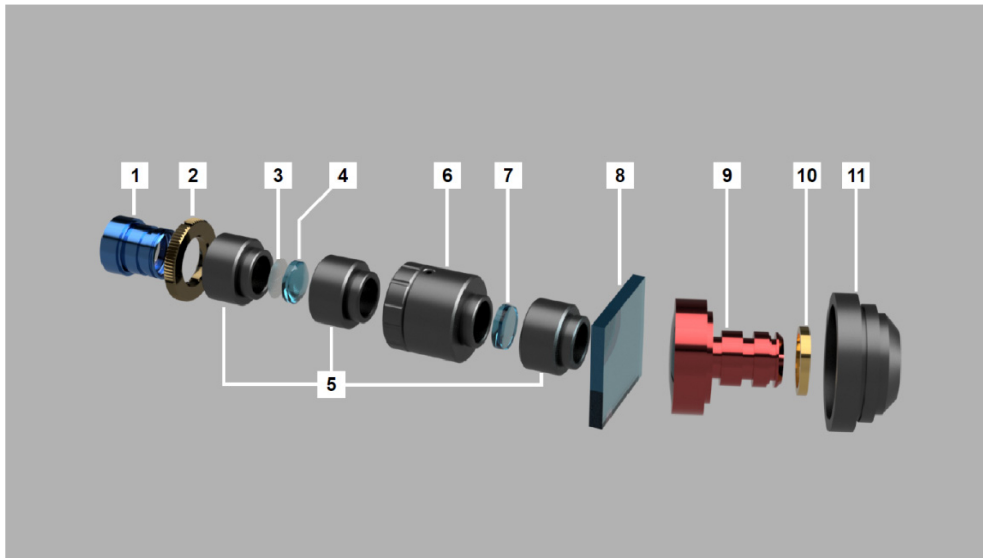


Figure 1: Exploded view of Hyper Spectral Imager (HSI) components: (1) front lens, (2) lock nut, (3) air slit, (4) field lens, (5) three thin lens mounts, (6) focus tube, (7) collimator lens, (8) transmission grating, (9) detector lens, (10) focus spacers, and (11) C-mount lens adapter. More information in Sigernes et al. 2018.

2 Tasks in the lab

This section provides you with a description of how to operate the optics lab and what you need to do while there. A how to guide Spectral Calibration is given in section 4. A how to guide for the Radiometric Calibration is given in section 5.

You will be allocated a 3 hour time-slot to perform the characterization, but with good preparation, the characterization can be completed under 1.5 hours. But if necessary, we can accommodate extra sessions if your allocated time-slot is not enough. Contact the lab assistant for scheduling extra lab sessions.

Each characterization involves taking images with the instruments under a certain lab setup. After completing the data collection, you would ideally have a set of images similar to what can be found in table 1. This a recommendation, and you are free to adjust the number of camera and target configurations and samples of each, if you are curious. The recommended number of images will be enough to complete the assignment.

The column titled **N** in Table 1 indicates the recommended configurations times the number of images per configuration.

Table 1: A table showing what you should collect during the lab session.

Type of Image	Justification & Use	N
Spectral Calibration Lamp Images	For Spectral Calibration and characterization. Make sure that the images are not overexposed. Take 10 images with the first lamp and 10 images with the second lamp following the guide given in section 4.	2 x 10
Radiometric Calibration Screen Images	For radiometric calibration and characterization. Make sure that the images are not overexposed. Take 10 images for three different exposure times following the guide given in section 5.	3 x 10
Dark Images	To estimate dark current noise. Take 10 images using the same exposure times as in the radiometric sampling in complete darkness.	3 x 10

3 Tasks

The following subsections detail tasks and objectives related to the assignment. Attempt to complete them all. The steps you need to include in your report **are all in bold**.

3.1 Calibration

This subsection describes tasks and objectives related to the spectral and radiometric calibration of the HSI in this assignment. The data from spectral calibration should be used to measure the FWHM in section 3.2.1, while the radiometric calibration data should be used to assess the sensor response. The main equipment that is needed to collect the needed data is a halogen lamp and calibration screen, as detailed in section 5. **Describe the calibration instruments. Describe the procedures needed to collect the desired data for spectral and radiometric calibration.**

3.1.1 Spectral Calibration

For the spectral calibration, i.e. the wavelength-pixel relation, you will need to perform some interpolation to fit to the data. In this assignment, you will perform spectral calibration using data from spectral lamps and a calibration screen. **To complete the task justify an approach you would have selected with respect to the following questions:**

- **What type of spectral lamps will you use, and why?**
- **How many images will you take for each lamp?**
- **What criteria will you use to determine the exposure time?**

Create a combined image containing emission lines from all, or at least two lamp sources (e.g. by addition), and comment.

It is common to use the spectral pixel position of each emission peak and compare them to the known positions of the elemental emission peaks to estimate the spectral calibration, by e.g. interpolation. **Present the Root Mean Square Error (RMSE) of a set of interpolation or model fitting methods, e.g. polynomial fits of various degrees, used to map the pixel position to wavelength and comment on the results. What kind of method or model seem to best describe the relationship? Comment on the selected methods.**

3.1.2 Radiometric Calibration

The radiometric calibration data is to be collected using a halogen lamp and a calibration screen where the HSI was placed in front of the screen. This is shown in section 5.

Convert each collected frame S to units of [counts/sec], using the frame's exposure time. Average all frames with same exposure time. Remember to subtract corresponding dark frames prior to the operation. This average image will help to reduce random fluctuations, e.g. noise of various kinds. The radiometric calibration coefficient for each pixel, K , can be seen as a scaling factor and can be calculated as $K = L/S$, where L is given in eq. (1).

$$L = L_0 \times (r/R)^2 \times \cos(\alpha) \times \rho \quad (1)$$

The quantity L_0 is the certificate irradiance in $\text{mW}/(\text{m}^2 \text{ nm})$, α is the tilt angle between the center of the screen and the lamp, r is the certificate distance. R is the distance from lamp to center of

screen in the lab, and ρ is diffuse emittance coefficient of the screen (in the lab 0.98). The lamp should be in the center, head on illuminated, which means $\alpha = 0$.

The angle ϕ of the center of the screen and instrument does not matter as long as the whole field of view of the HSI is inside the illuminated area of the screen. You will need to measure R .

K is different for each pixel, and therefore also should correct for any inter-pixel variations. **Visualize a matrix of all the K -values. Comment on the results.**

The uncertainty of radiometric calibration coefficients can be estimated by radiometrically calibrating a single frame, subtracting it from a reference frame, and dividing the difference by the reference values. The mean or median image of several images can be used a reference frame. **Plot the uncertainty of the radiometric calibration coefficients, and comment on the results.**¹.²

¹Which spatial line do you look at and why?

²What effects do you expect from using the median image versus the mean image as a reference?

3.2 Characterization

The following subsections detail tasks and objectives related to sensor characterization.

3.2.1 Spectral resolvability

The spectral full-width-half-maximum (FWHM) is a number quantifying spectral resolution. Compute the theoretical FWHM using the slit width, the grating groove spacing, and the focal length of the middle objective that collimates the light onto the grating. **Compute the theoretical FWHM for the first three orders, and comment on the result.** You can find the necessary specification in Sigernes et al. 2018. The first spectral order is what will be relevant the rest of the assignment.

Empirically, the FWHM can be estimated using using images of the light emission spectra from a mercury and/or argon lamp. This will form the basis for the spectral calibration data as well. The peaks should be identified, and the width at half maximum of should be found for several positions along the spatial axis. Keep in mind that some of the detected peaks are known to be double peaks (two peaks lying very close to each other) and should be excluded. **Plot the FWHM for various wavelengths as a function of the spatial axis, and comment on the results.**

3.2.2 Radiometric response

The data from radiometric calibration can be used to investigate the radiometric response.

A vignetting effect is visible, making the center of the image brighter (meaning a stronger signal) and the edges darker. Dust on the slit can form horizontal stripes.

The absolute radiometric response can be computed as the average pixel response for each wavelength by taking the mean of the columns in the radiometric calibration image.

The smile effect can be ignored for this assignment as the absolute radiometric response shall only be used for visual inspection. **Plot a normalized absolute radiometric response of the spatial center line of the image alongside the radiometric response of the calibration screen and comment on the results.**

Examine how the response varies with exposure time. Knowing the sensor response as a function of exposure time enables you to adjust the exposure according to the image target. Investigation of the sensor response can be done using the set of images of the calibration screen captured at different exposure times. **Describe the procedure to collect the sensor response and the response itself and comment on when the results will hold.**

4 Spectral Calibration

How-to: Spectral calibration in the optics lab

This document entails detailed information regarding how to use the equipment in the optics lab to perform spectral calibration for a hyperspectral imager.

1. Take a picture of the lab.
2. Find all the necessary equipment.
This includes:
 - Power supply for spectral tubes (Part 1)
 - Spectral tubes (Part 2) and plastic gloves
 - Protective glasses (Part 3)
 - Hyperspectral imager (Part 4)
 - 3D-printed disc with a tape filament (Part 5)
 - Computer with applicable software (Part 6)
 - iOptron rotating stage (Part 7)
 - Machined part between rotating stage and camera platform (Part 8)
 - 3D-printed part keep rotating stage from tilting over when weighted (Part 9)

The power supply with a spectral tube is shown in Figure 1.

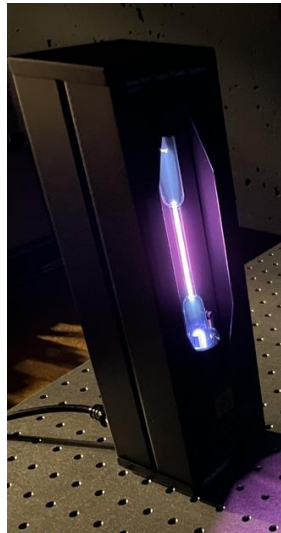


Figure 1: Power supply (Part 1) and spectral tube (Part2).

The rotating stage is shown in Figure 2.



Figure 2: iOptron rotating stage (Part 7).

A machined part is necessary to fasten the camera to the iOptron rotating stage. This is shown in Figure 3.



Figure 3: Machined part (Part 8)

The 3D-printed part that keeps the rotating stage from tilting (Part 9) looks like bracket as is first fastened to the optics table. After fastened, the rotating stage is slid underneath the bracket (Could be helpful with an image).

An example set-up of the test is shown below in Figure 4.

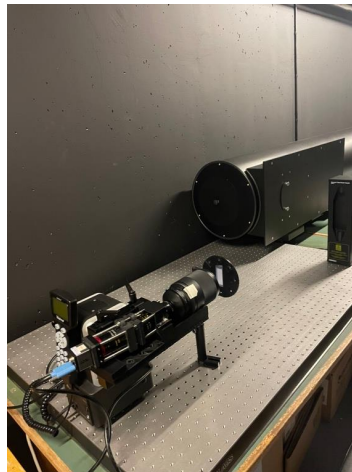


Figure 4: Example of set-up.

In Figure 4, the HSI v6 is used. If the v6 is used, a 3D-printed platform (Part 4.1) is necessary to fasten the camera to the rotating stage.

The 3D-printed disc with the tape filament is supposed to be between the camera and spectral tube and can be seen in Figure 4. This is to fill the whole slit with light (you will hopefully see the lines widen when using this disc). When using the disc, if you tilt the light source left and right, the spectral lines should not move. The tape, however, hinders some of the light to reach the camera. Only if necessary, increase the gain to have enough signal.

3. Set-up:

- a. Fasten Part 9 to the optics table using two M6 screws (found in the lab).
- b. Slide the iOptron rotating stage (Part 7) under Part 9. Make sure the rotating stage is secured.
- c. Fasten the imager (Part 4) onto Part 8. Use Part 4.1 if necessary.
- d. Secure Part 8 onto the rotating stage (Part 7).
- e. Align the camera (Part 4) and the power supply (Part 1). If the camera has a small FOV some adjusting might be necessary before finally taking images.
You might have to adjust the angle of the rotating stage. If so; turn it on, click on “Back” when asked to calibrate, and adjust the orientation via the control. Adjust speed by clicking on numbers 1-9.
- f. Place Part 5 between the camera and the power supply. Again, some adjusting before taking images may be required to ensure that the light passes through the tape filament onto the camera.
- g. Connect the camera (Part 4) to the computer (Part 6) and open the camera software.
- h. Place one of the spectral tubes (Part 2) in the power supply (Part 1).
Remember to use plastic gloves, when touching the spectral tubes.

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- i. Turn off the overhead lights in the lab
 - j. Put the protective glasses on
 - k. Turn on the power supply. Do not put max power as this can damage some of the spectral tubes.
 - l. Adjust camera settings. Only use gain if necessary. Adjustment of the position of some equipment might be necessary.
 4. Take images. Take at least 10 for each lamp. Remember to use plastic gloves when switching tubes.
 5. Clean up all equipment.
 6. Take a picture of the lab and compare it to the picture in step 1.

5 Radiometric Calibration

How-to: Radiometric calibration in the optics lab

This document entails detailed information regarding how to use the equipment in the optics lab to perform radiometric calibration for a hyperspectral imager.

1. Take a picture of the lab.
2. Find all the necessary equipment.
This includes:
 - Power supply for halogen lamp (Part 1)
 - Halogen lamp (Part 2)
 - Long extension cable (Part 3)
 - Hyperspectral imager (Part 4)
 - Computer with Remote Desktop installed (Part 5)
 - iOptron rotating stage (Part 6)
 - Machined part between rotating stage and camera platform (Part 7)
 - 3D-printed part keep rotating stage from tilting over when weighted (Part 8)

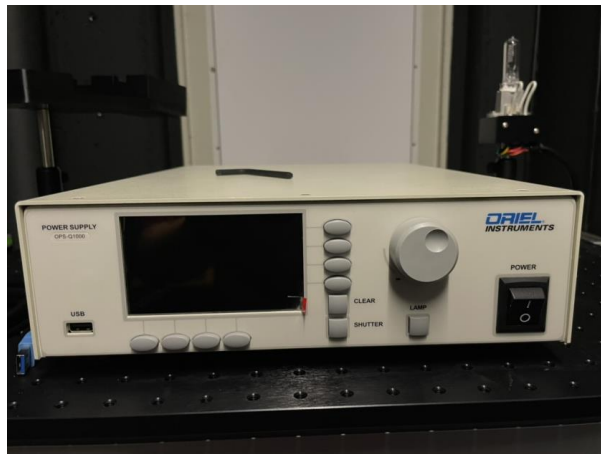


Figure 1: Power supply (Part 1) for the halogen lamp (Part 2). The halogen lamp is seen in the top right corner of the figure.



Figure 2: iOptron rotating stage (Part 6).



Figure 3: Machined part (Part 7).

The 3D-printed part that keeps the rotating stage from tilting (Part 8) looks like bracket as is first fastened to the optics table. After fastened, the rotating stage is slid underneath the bracket (Could be helpful with an image).

An example set-up is seen in Figure 4.

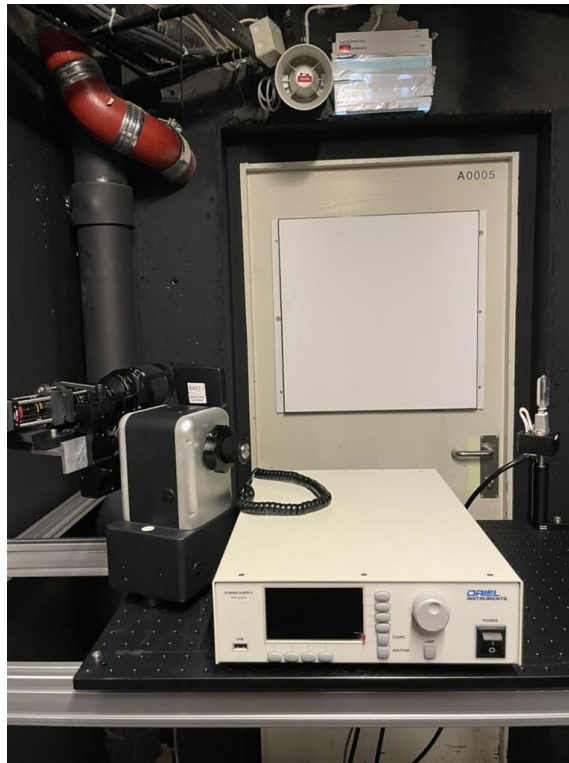


Figure 4: Set-up of radiometric calibration. The HSI v6 is mounted on the iOptron rotatin stage (Part 6).

In Figure 4, the HSI v6 is used. If the v6 is used, a 3D-printed platform (Part 4.1) is necessary to fasten the camera to the rotating stage.

3. Set-up:

- a. Fasten Part 8 to the optics table using two M6 screws (found in the lab).
- b. Slide the iOptron rotating stage (Part 6) under Part 8. Make sure the rotating stage is secured.
- c. Fasten the imager (Part 4) onto Part 7. Use Part 4.1 if necessary.
- d. Secure Part 7 onto the rotating stage (Part 6).
- e. Turn on the rotating stage. Click “Back” when asked to calibrate.
- f. Adjust the orientation of the rotating stage such that the camera points 90 degrees on the wall of the calibration screen. Navigate to Zero Position on the control remote of the rotating stage and click on “Set Zero Position”.
- g. Adjust only the azimuth angle of the rotating stage until the camera points at the calibration screen. Note down the change in azimuth angle from the zero position (zero position should have azimuth angle of 180 degrees). This angle is α in Figure 5 and is necessary to do the calculations correctly.

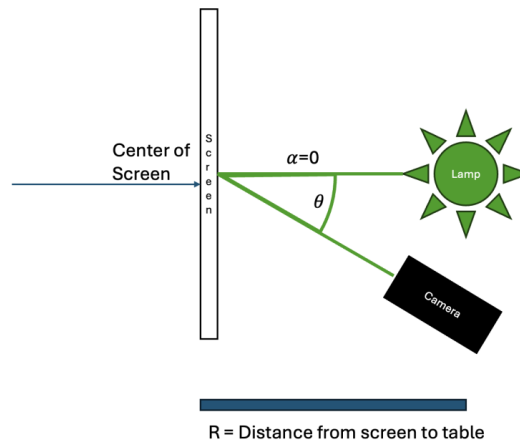


Figure 5: Sketch of set-up at lab.

- h. Plug the extension cable in an outlet outside of the lab and bring it next to the power supply. Plug the power supply's power cable into the extension cable.
- i. Plug the USB cables from the power supply and camera into the lab computer (there is a hub underneath the optics table the set-up rests on).
- j. Turn on the power supply, and make sure the lamp is connected to the power supply.
- k. Click on button "1" and then "2" as seen in Figure 6. Adjust to wanted power by rotating the knob. The power sent to the lamp is "P-set".

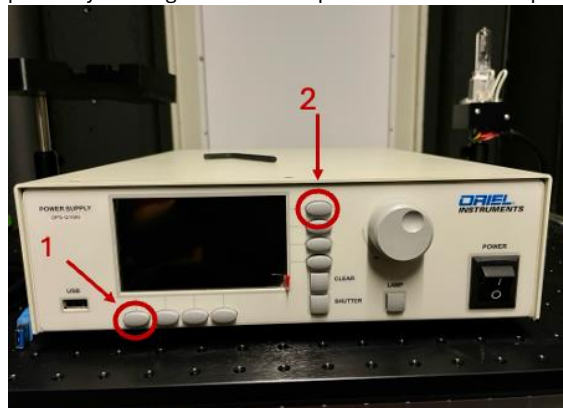


Figure 6: Power supply buttons.

- l. Navigate to the ODevice.vi on the lab computer and open it in the LabVIEW program.
- m. Open your PC and connect to the lab computer using remote desktop. ALL NEXT STEPS ARE FROM YOUR COMPUTER.

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- n. Close the lights, step outside of the lab and close the door.
4. From the drop-down bar in LabVIEW, choose “open” and click on the run-button in the LabVIEW interface. The power supply should be connected and should print this in the print part of the LabVIEW program.
 5. Make sure you have the wanted power setting by choosing “query”, write “p-set?” and click on the run-button.
 6. Open the camera software on the lab computer (via your computer).
 7. In LabVIEW, from the drop-down bar, choose “write”, write “start” and click on the run-button. You should now see the light turning on via the camera software. You can verify the current power to the lamp by choosing “query” and writing “watts?” in LabVIEW.
 8. Adjust the camera settings. Preferably without any gain.
 9. Take at least ten images for the current setting. Feel free to try different exposure times.
 10. In LabVIEW, from the drop-down bar, choose “write”, write “stop” and click on the run-button. You should now see the light turning off via the camera software. You can verify the current power to the lamp by choosing “query” and writing “watts?” in LabVIEW. Do not go in before you see the lamp power is 0.
 11. Clean up all equipment.
 12. Take a picture of the lab and compare it to the picture in step 1.

Bibliography

- Henriksen, Marie Bøe, Elizabeth Frances Prentice et al. (2022). ‘Pre-Launch Calibration of the HYPSO-1 Cubesat Hyperspectral Imager’. In: *2022 IEEE Aerospace Conference (AERO)*, pp. 1–9. DOI: 10.1109/AERO53065.2022.9843655.
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- Sigernes, Fred et al. (Mar. 2018). ‘Do it yourself hyperspectral imager for handheld to airborne operations’. In: *Opt. Express* 26.5, pp. 6021–6035. DOI: 10.1364/OE.26.006021. URL: <https://opg.optica.org/oe/abstract.cfm?URI=oe-26-5-6021>.